A CERTIFIED SQUARE PAPER TORUS WITH EIGHT VERTICES

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Abstract. The most symmetric flat tori are the square torus and the hexagonal torus, the two orbifold points of the moduli space. We realize both as embedded polyhedral tori in $\mathbb{R}^3$ with 8 vertices, the minimum possible by Schwartz's bound. Both lie on the locus that the near-universality theorem of Doyle and Schwartz excludes, the rectangular tori together with the rhombic tori of aspect ratio at least $\sqrt{3}$, and no vertex-minimal realization of either was known. The proof is computer-assisted and fully validated: a Krawczyk interval certificate proves that a flat configuration of modulus exactly i exists in an explicit coordinate box, a separating-axis certificate in exact rational arithmetic proves that every configuration in that box is embedded, and the hexagonal torus is certified in the same way on its own triangulation. The certificates rely on no library transcendental function and are reproduced, for both tori, in two further independent interval arithmetic implementations.

1. Introduction

Take a flat sheet of paper, and fold and glue it into a torus. The intrinsic geometry of the sheet never changes, and the result is a *flat torus*, a surface with zero Gaussian curvature everywhere, realized in three-space as a polyhedral surface. Such *paper tori* were first constructed by Burago and Zalgaller [1], who later proved that every isometry class of flat torus is realized, though only by constructions of enormous size [2] (see the discussion in [8]). Explicit constructions followed, from Brehm's diplotori [5] to origami tori [3] and the diplotorus family of Arnoux, Lelièvre, and Málaga [4]. Quintanar embedded the square torus explicitly [6], and Lazarus and Tallerie built a single universal triangulation, with 2434 faces, admitting every flat torus [8]. The modern question is how few vertices a paper torus can have. Tugayé found a 9-vertex paper torus [9]. Schwartz then answered the question completely, proving that no paper torus has 7 vertices and that 8-vertex paper tori exist [10], so eight is the minimum possible.

Which shapes of flat torus arise with eight vertices? The shape is captured by a single point $\hat{\tau}$ in the classical modular domain (Section 2). Doyle and Schwartz proved near universality, that every flat torus *without reflection symmetry* is an 8-vertex paper torus [11]. The pup tent [10] carries an order-2 rotational symmetry, and its moduli sweep out the interior of the bi-cusped fundamental domain of [11]. In the coordinates of Section 2 these are the $\hat{\tau}$ with $0 < |\operatorname{Re}\hat{\tau}| < \frac{1}{2}$, including the rhombic tori of aspect ratio in $(1, \sqrt{3})$, the open arc of $|\hat{\tau}| = 1$ between the orbifold points. Excluded are exactly the rectangular tori, the set $\{\, iY \mid Y \geq 1 \,\}$, and the rhombic tori of aspect ratio at least $\sqrt{3}$, the set $\{\, \pm\frac{1}{2} + iy \mid y \geq \frac{\sqrt{3}}{2} \,\}$. Doyle and Schwartz ask whether these excluded shapes can be realized if one drops the symmetry of the pup tent.

This paper answers that question for the two most symmetric points, namely the square torus $\hat{\tau} = i$, realized on the same triangulation as the pup tents, and the hexagonal torus $\hat{\tau} = e^{i\pi/3}$, realized on a second triangulation. They are the two orbifold points of the moduli space, sitting at the corners of the excluded locus. Doyle and Schwartz come arbitrarily close, but their construction breaks down at the boundary. Their collapsing families contain 8-vertex paper tori

Date: August 13, 2026.

whose moduli converge to the square and hexagonal points, while the tori themselves degenerate, their vertices colliding in the limit. The square torus had been embedded polyhedrally before, by Quintanar with 40 vertices [6]. Our result is that eight suffice,[1] the fewest possible, by Schwartz's lower bound [10].

Theorem 1. *There exists an embedded flat polyhedral torus in $\mathbb{R}^3$ with 8 vertices, 24 edges, and 16 faces, combinatorially the triangulation T of Figure 4, whose intrinsic flat metric is that of the square torus $\mathbb{R}^2/\mathbb{Z}^2$. Since no torus triangulation has fewer than 7 vertices and no paper torus has 7 [10], it is vertex-minimal.*

The triangulation T is the same as in [10], the unique 8-vertex torus triangulation in which every vertex has valence 6, the type Schwartz's pup tents use (Figure 1). The pup tents' order-2 rotation does not pin the modulus, and they sweep out an open set of moduli, the interior of the bi-cusped domain of [11]. The square torus lies on the excluded rectangular ray, outside that open set. Our realization uses the same triangulation but no spatial symmetry.

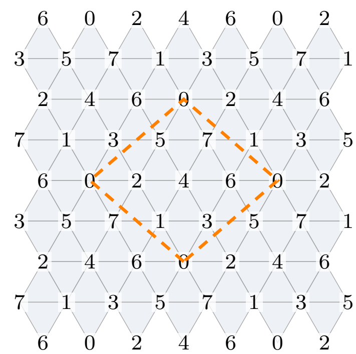

FIGURE 1. The triangulation T, shown through its universal cover, the regular $\{3, 6\}$ triangulation of the plane. Every vertex has valence 6. One fundamental domain is outlined, with 8 vertices, labeled 0 through 7, and 16 triangles.

Remark 1. We would like to say that Theorem 1 means exactly that the square torus can be folded from a square sheet of paper along the edges of T. Whether embeddability of the polyhedral surface is equivalent to foldability in that hands-on sense is a subtle question that we do not settle. We can report the experimental fact that this particular configuration does fold up in $\mathbb{R}^3$.

We also found a numerical configuration for the hexagonal torus, the other orbifold point.

Theorem 2. *On a second 8-vertex triangulation, with degree sequence $(4, 6, 6, 6, 6, 6, 7, 7)$ rather than the valence-regular T, there exists an embedded flat polyhedral torus in $\mathbb{R}^3$ with 8 vertices, 24 edges, and 16 faces whose intrinsic flat metric is that of the hexagonal torus $\mathbb{C}/(\mathbb{Z} \oplus e^{i\pi/3}\mathbb{Z})$, reduced representative $\hat{\tau} = e^{i\pi/3}$.*

The proof of Theorem 1, run on the second triangulation with the modulus pinned to $e^{i\pi/3}$, succeeds unchanged. It is carried out in Appendix C.

A numerical search through configurations of 8 points in $\mathbb{R}^3$ produces a candidate, an embedded configuration that is flat to 10^{-15} with modulus i to 10^{-15}. Flatness and modulus

[1] A flat square torus on 8 vertices appears earlier in [7], but it is degenerate, of zero volume with coincident faces, and not embedded.

together are 9 analytic conditions on the 24 coordinates and freezing 15 coordinates leaves a square 9×9 system. The Krawczyk interval test, run in outward-rounded double-precision interval arithmetic, proves that this system has exactly one true zero in an explicit box of radius 10^{-9} around the candidate, i.e. an exactly flat torus of modulus exactly i. A second certificate, run in exact rational arithmetic, proves that every configuration in that box is embedded, using separating axes for the disjoint and shared-vertex pairs, one-signed tetrahedral volumes for the edge-shared pairs, and two general-position certificates. The zero lands in the box, so it is embedded, proving the theorem.

Every computational claim in the proof is a machine-verified certificate, an interval computation whose soundness rests only on the IEEE-754 behavior of $+, -, \times, \div, \sqrt{}$ in double precision, with every transcendental built from scratch and self-verified (Section 6). Floats enter a certificate only as initial data for its hypothesis, centers and preconditioners. The hypothesis is verified using interval arithmetic, and the conclusion never mentions them. The certificates rerun end to end in about six seconds (Appendix A) and are cross-validated by two further independent interval implementations (mpmath and Arb) and an exact-rational rerun of the embedding. The lemmas connecting the certificates to the theorem are proved in full below.

The method is similar to Schwartz's [10], with three changes. He proves existence of the pup tents by a high-precision solution, a square system of cone-angle equations, and an effective inverse function theorem whose constants are proved by hand from a high-precision residual and Jacobian [10]. We work on the same triangulation but without the symmetry that makes his system 3×3, paying with a 9×9 system. We pin the modulus to the point i with two explicit equations, because i lies on a measure-zero locus that no open-set argument can reach. And we replace the floating-point evaluations by a fully validated Krawczyk certificate, including an embeddedness proof over the whole box in exact rational arithmetic. Section 7 makes the comparison precise.

Outline. Section 2 defines flat and paper tori, their moduli, and explains why the square point demands special coordinates. Section 3 presents the triangulation and the certified configuration. Section 4 proves existence of the exact flat torus (the Krawczyk certificate). Section 5 proves it is embedded (the separating-axis certificates and the two geometric lemmas behind them). Section 6 is the certification model, including what the computation assumes and the independent cross-validations. Section 7 details the relation to Schwartz's argument, and Section 8 records experimental findings and open questions. Appendices: the certificate table with reproduction instructions, an exactly-embedded rational witness which is numerically close to being flat, and the hexagonal companion.

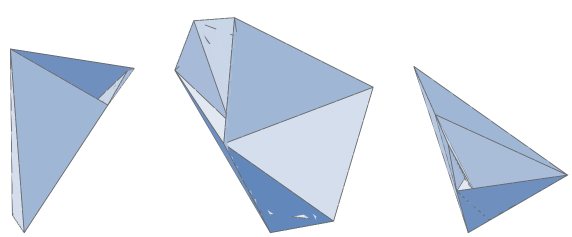

FIGURE 2. The certified square paper torus from three angles, orthographic projections of the configuration of Table 1, faces painter-sorted and shaded by orientation to the light. In the rightmost view the line of sight passes through the hole of the torus. Every configuration in a box of radius 10^{-9} around it is embedded (Theorem 5), and the exactly flat, exactly square torus is one of them (Theorem 4).

Acknowledgements. The example came out of code written jointly with Steve Trettel in early June 2026, as a gift for Richard Schwartz's birthday conference, made in the name of Anna Wienhard's group at MPI MiS Leipzig. It grew out of many conversations with Steve, from which I learned a great deal, and I am grateful to him for his generosity and insight throughout. I thank Richard Schwartz for his encouragement to write up this result.

2. Flat tori, paper tori, and their moduli

2.1. **Lattices and the modulus.** A *flat torus* is the quotient $\mathbb{C}/\Lambda$ of the plane by a lattice $\Lambda = \mathbb{Z}v_1 \oplus \mathbb{Z}v_2$. Rotating and scaling the plane changes (v_1, v_2) to $(\lambda v_1, \lambda v_2)$ without changing the intrinsic shape, so the shape is captured by the ratio $\tau = v_2/v_1$ in the upper half plane $\mathbb{H}$ and changing the basis of Λ moves τ by an element of the modular group $\mathrm{PSL}(2,\mathbb{Z})$, via Möbius transformations $\tau \mapsto \frac{a\tau+b}{c\tau+d}$. Every orbit meets the closed fundamental domain $\mathcal{F} = \{\, |\tau| \geq 1,\ |\operatorname{Re}\tau| \leq \frac{1}{2}\,\}$, and its representative there is unique once the boundary of $\mathcal{F}$ is glued, the two vertical sides $\operatorname{Re}\tau = \pm\frac{1}{2}$ by $\tau \mapsto \tau+1$, and the two halves of the arc $|\tau| = 1$ by $S\colon \tau \mapsto -1/\tau$ (Figure 3). We call this point of the resulting modular surface the *reduced representative* $\hat\tau$ (see [14, §12.2] for a thorough treatment). The tori with an orientation-reversing (reflection) symmetry are exactly the *rectangular* tori $\hat\tau = iY$, $Y \geq 1$, and the rhombic tori. We are interested in $\hat\tau = i$, the orbifold point where the rectangular and rhombic loci meet, the square torus $\mathbb{C}/(\mathbb{Z} \oplus i\mathbb{Z}) = \mathbb{R}^2/\mathbb{Z}^2$ (Figure 3).

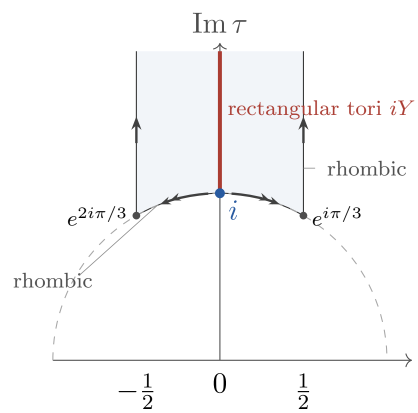

Figure 3. The fundamental domain $\mathcal{F}$ of the modular group. The pup tents of [11] realize every modulus off the rectangular line $\operatorname{Re}\hat\tau = 0$ and the rhombic sides $\operatorname{Re}\hat\tau = \pm\frac{1}{2}$, the open arc between i and $e^{i\pi/3}$ included. This paper realizes the orbifold point $\hat\tau = i$, the square torus, where the rectangular tori iY $(Y \geq 1)$ meet the rhombic arc. The arrows are the identifications that fold $\mathcal{F}$ into the modular surface ($\tau \mapsto \tau+1$ glues the sides, S folds the arc at i). The rhombic tori are the arc and sides, and $e^{\pm i\pi/3}$ is the order-3 hexagonal point.

2.2. **Paper tori and flatness.** Fix a triangulation T of the torus with vertex set $V = \{v_0, \ldots, v_7\}$. Up to combinatorial isomorphism there are exactly seven such T [12, 13], and Section 3 singles out the one we use. Since the torus has Euler characteristic zero and every face is a triangle, any such T has 24 edges and 16 faces. A *configuration* is a continuous map $c\colon T \to \mathbb{R}^3$, affine on each face, fully determined by the images $c(v) \in \mathbb{R}^3$ of the eight vertices. We identify c with the point $(c(v_0), \ldots, c(v_7)) \in \mathbb{R}^{24}$. It is *flat* if the cone angle at every vertex equals 2π, that is, if the sum of the incident triangle angles equals 2π. The intrinsic metric then has no cone points, the surface develops onto the plane, and it is a flat torus $\mathbb{C}/\Lambda$ with a well-defined modulus. A configuration that is both flat and injective is a *paper torus*, an embedded flat polyhedral torus. We write $\delta_v = 2\pi - \theta_v$ for the *deficit* at v, with θ_v the cone angle. On a torus the deficits always sum to zero:

$$\begin{aligned}\sum_v \delta_v = \sum_v (2\pi - \theta_v) &= (\#\text{vertices}) \cdot 2\pi - \sum_v \theta_v \\ &= (\#\text{vertices}) \cdot 2\pi - (\#\text{faces}) \cdot \pi = 8 \cdot 2\pi - 16 \cdot \pi = 0,\end{aligned}$$

the polyhedral Gauss–Bonnet identity. So flatness is 7 conditions, not 8. This identity holds for every configuration with nondegenerate faces, flat or not. The map we certify depends only on the deficits $\delta_0, \ldots, \delta_6$, and if they are 0, then $\delta_7 = 0$ automatically.

For a flat configuration the modulus is read off the *developing map*. Unfold the triangles into the plane, one after the other across shared edges. Going around a closed loop of the torus translates the developed picture by a period vector, and two independent loops give two periods v_1, v_2 with $\tau = v_2/v_1$. Fixing the order in which the triangles are unfolded and a choice of two such loops (a *marking*) makes τ a well-defined smooth function of the configuration. Our marking data, two explicit edge loops on T, is shown in Figure 5. It is a homology basis (Section 3).

2.3. **The square point is an orbifold point.** The square point i is an orbifold point of the moduli space $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$, so we work with the raw modulus τ, which for our fixed marking is a smooth function of the configuration, and seek a zero of $\tau - i$,

$$\tau_{\mathrm{re}}(c) = 0, \qquad \tau_{\mathrm{im}}(c) = 1.$$

A zero has marked modulus i, the square torus. Any representative of the orbit $\mathrm{PSL}(2,\mathbb{Z}) \cdot i$ would have served equally.

2.4. **Notation.**

T, V	the triangulation of Figure 4 and its 8 vertices
$c = (P_0, \ldots, P_7) \in \mathbb{R}^{24}$	a configuration; $P_v \in \mathbb{R}^3$ the position of vertex v
δ_v	the angle deficit at vertex v
τ, $\hat{\tau}$	the marked (raw) modulus; the reduced representative
$G\colon \mathbb{R}^{24} \to \mathbb{R}^9$	the constraint map $(\delta_0, \ldots, \delta_6, \tau_{\mathrm{re}}, \tau_{\mathrm{im}} - 1)$
$F\colon \mathbb{R}^9 \to \mathbb{R}^9$	G restricted to the 9 free coordinates (Section 4)
$\check{c}$, $\check{x}$	the certified center (Table 1); its nine free coordinates
r, X	the box radius; the box $X = \prod_{j=1}^{9} [\, \check{x}_j - r,\ \check{x}_j + r\,]$

3. The certified object

3.1. **The triangulation.** T has 8 vertices, 24 edges and 16 faces. Its faces are

356, 325, 364, 302, 341, 310, 506, 524,

547, 570, 674, 601, 617, 214, 207, 271.

Among the seven combinatorial types of 8-vertex torus triangulation, enumerated by Sulanke and Lutz [12, 13], T is the unique one in which every vertex has valence 6 (Figure 1). It is the triangulation we work with, the pup-tent type of Schwartz [10] and Doyle–Schwartz [11]. The hexagonal companion of Theorem 2 is realized on a different 8-vertex triangulation, given in Appendix C. Figure 4 shows T through the development of the certified configuration.

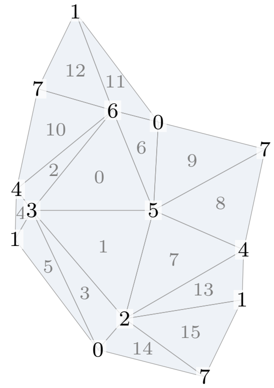

FIGURE 4. The triangulation T, shown as the developed net of the certified configuration, sixteen triangles, faces numbered in slot order, vertex labels repeating where edges are glued.

3.2. **The certified configuration.** The certified object is an explicit list of 24 numbers, the vertex coordinates of Table 1, which were found by a numerical search. Here is the story in short. A rough embedded candidate near the square modulus was refined by alternating Gauss–Newton projection onto the flat, square-modulus locus $\{\delta = 0\} \cap \{\tau = (0, 1)\}$ with gradient descent of an embedding energy along it, until the residual reached 10^{-15}. Nothing in the proof depends on any of this. The search produced a candidate, and the certificates of Sections 4 and 5 take it as their center.

v	x_v	y_v	z_v
0	−0.32228466828574437	0.4637721192185657	−0.74323457006951088
1	0.2324200435776736	−0.46217058403057881	0.24599012355582775
2	−0.14382327673249617	0.079722048364084933	−0.83230338087592548
3	0.22779161128071213	−0.32944176958982457	0.57668754870786809
4	0.17876493830658916	−0.057582559083758984	0.58547474269171451
5	−0.12429063916040699	0.81747961438065686	0.10086387909811872
6	−0.24593729785582902	−0.023388654255220555	−0.66463839618166654
7	−0.62529227417585398	−0.42715999499773932	−0.049686468538500325

TABLE 1. The certified center $\check{c}$, the eight vertex positions, to full double precision. The table is from the data file `code/data/square-i-type7.csv`, which is the exact input of every certificate. The exactly flat, exactly square torus of Theorem 4 lies within 10^{-9} of it, coordinate by coordinate. The listed configuration itself, the center of the box X, is embedded by Theorem 5, and exactly embedded in rational arithmetic (Appendix B).

At the center, the interval certificates of Section 6 enclose the seven deficits within 1.93×10^{-13} of zero and the marked modulus within 2×10^{-13} of i (and sharper evaluations place both within 10^{-15}). The configuration carries no spatial symmetry, more precisely no isometry of $\mathbb{R}^3$ permutes its vertices. Figure 5 shows the developed net again, now with the two marking loops and their holonomy vectors and the period lattice of the certified metric is visibly square.

3.3. **The marking is a homology basis.** The two marking loops of Figure 5 are simple closed curves, the red loop $4 \to 2 \to 3 \to 4$ and the blue loop $5 \to 2 \to 0 \to 5$, and they meet only at vertex 2. The six faces incident to that vertex are $325, 302, 524, 214, 207, 271$, so its neighbors close up into the link cycle $(3, 5, 4, 1, 7, 0)$. The red loop leaves vertex 2 along the edges to 4 and 3, the blue along the edges to 5 and 0, and in the link cycle these two pairs alternate. The two loops therefore cross exactly once, with algebraic intersection number ± 1. The intersection pairing on $H_1(T; \mathbb{Z}) \cong \mathbb{Z}^2$ is unimodular, so any two classes with intersection number ± 1 form a basis, and our two loops are such a pair.

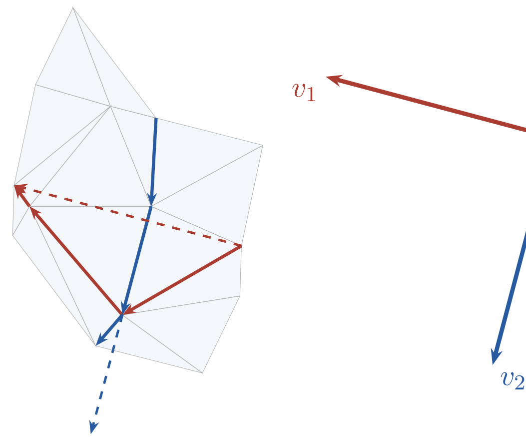

FIGURE 5. The development of the certified configuration with the two marking loops (red for the loop $4 \to 2 \to 3 \to 4$, blue for $5 \to 2 \to 0 \to 5$). Each loop edge is drawn in one developed triangle containing it, and the holonomy vector of a loop is the sum of its developed edge vectors, drawn dashed from the loop's starting vertex. The inset repeats the two holonomy vectors from a common origin, to the same scale. v_2 is v_1 rotated by $90°$ to 15 digits.

4. EXISTENCE VIA THE KRAWCZYK CERTIFICATE

4.1. **The constraint map and the square slice.** Assemble the nine conditions into one map,

$$G\colon \mathbb{R}^{24} \to \mathbb{R}^9, \qquad G(c) = \big(\delta_0, \ldots, \delta_6,\ \tau_{\mathrm{re}},\ \tau_{\mathrm{im}} - 1\big). \tag{1}$$

By Gauss–Bonnet (Section 2), a zero of G is flat at all eight vertices and has marked modulus exactly i. A zero of G that is also embedded is exactly what Theorem 1 asks for.

The zero set of G is never isolated because G is invariant under the 7-dimensional group of orientation-preserving similarities of $\mathbb{R}^3$ (three translations, three rotations, one scaling), since translating, rotating, or rescaling a configuration changes no cone angle and leaves the modulus fixed. Every zero thus comes with at least the 7-dimensional similarity orbit, so none is isolated. Numerics suggest that G is a submersion, in which case a generic 9×9 coordinate slice of G has an isolated zero, which is what the Krawczyk machinery needs. So we pass to a *slice* by freezing 15 of the 24 coordinates at the center's values and letting the remaining 9 vary. On this slice G restricts to a square map,

$$F\colon \mathbb{R}^9 \to \mathbb{R}^9, \qquad F(y) = G\big(\text{center with the free coordinates replaced by } y\big). \tag{2}$$

Geometrically, on the slice the flat configurations form a surface with the modulus τ as a local coordinate, and the two equations $\tau = i$ cut out a single point of it. Which nine coordinates to free is purely a conditioning choice, made by column-pivoted Gram–Schmidt on the Jacobian of G at the center. It selects

$$\{\, z_3,\ z_0,\ y_1,\ y_5,\ z_6,\ z_4,\ x_7,\ z_5,\ y_0 \,\},$$

a mix of x, y, and z coordinates spread over seven of the eight vertices. No symmetry is imposed anywhere. The slice is well conditioned: at the center the 9×9 Jacobian slice has condition number[2] 94.1 (Appendix A row U1).

This method is a choice, not a necessity. The rest of this subsection records why the free-coordinate choice is robust, and a reader content to take that for granted may skip to the Krawczyk test. Enumerating all $\binom{24}{9} = 1{,}307{,}504$ slices confirms it. Most choices are well conditioned, meaning that the 9×9 slice has finite and modest condition number: it is below 10^3 for about 84% of the slices and below 10^4 for about 98%, and the best of all reaches $\kappa \approx 21$. The certificate is indifferent to which of these is used, since any slice of comparable conditioning works too. We use column-pivoted Gram–Schmidt rather than this exhaustive comparison because it is the method that scales and a reader facing a far larger system, where the slices cannot be enumerated, may try this constructive method first. The same enumeration turned up exactly 48 rank-deficient choices, and inspection showed these are precisely the ones that free an entire coordinate axis, all eight x-coordinates together, or all eight y, or all eight z. The reason is translation invariance. G is unchanged when every vertex is translated by a common vector, so $\sum_v \partial G/\partial x_v = 0$, that is, the eight x-columns of the Jacobian sum to zero (numerically to 10^{-15}), and likewise for y and z. These three sparse relations are the only kernel vectors of the Jacobian supported on nine or fewer coordinates, so a slice is singular exactly when it contains a full axis, which gives 16 completions of each axis times three axes. Column-pivoted Gram–Schmidt never selects such a slice. Once it has taken a few columns of an axis, the remaining same-axis columns have vanishing residual, so the rule skips them.

4.2. **The Krawczyk test.** The rigorous step replaces every estimate by an *enclosure*, a closed interval certified to contain the true value. Center the certificate at the nine free coordinates $\check{x} \in \mathbb{R}^9$ of the candidate, and take the box of radius r around it,

$$X \;=\; \check{x} \pm r \;=\; \prod_{j=1}^{9} [\,\check{x}_j - r,\; \check{x}_j + r\,] \;\subset\; \mathbb{R}^9,$$

a vector of nine intervals.[3] Let Y be a matrix approximating $F'(\check{x})^{-1}$, and let $F'(X)$ be an interval matrix enclosing the Jacobian of F over the whole box. The Krawczyk operator is

$$K(\check{x}, X) \;=\; \check{x} \;-\; Y\,F(\check{x}) \;+\; \big(I - Y\,F'(X)\big)(X - \check{x}).$$

Theorem 3 (Krawczyk–Moore)**.** *Suppose F is continuously differentiable on the box X and $F'(X)$ encloses its Jacobian there. For any $\check{x} \in X$ and any $Y \in \mathbb{R}^{n\times n}$, if the Krawczyk image $K(\check{x}, X)$ lies in the interior of X, then F has exactly one zero in X, and every matrix in $F'(X)$ is nonsingular.*

The operator is Krawczyk's [15] and that its interior containment certifies a *unique* zero goes back to Moore's test [16], in the sharpened form of [17, 18]. A good Y makes $I - YF'(X)$ small. So the operator contracts and the containment can hold. The theorem asks nothing of Y, though, any matrix is admissible, and a poor choice only makes the containment fail, never lets it certify a false zero.

[2]The condition number $\kappa = \sigma_{\max}/\sigma_{\min}$ measures how far a matrix is from singular. With $Y \approx F'(\check{x})^{-1}$ the Krawczyk contraction $\|I - YF'(X)\|$ scales as κ times the relative variation of the Jacobian across the box, so a small κ is what lets a box of radius 10^{-9} contract.

[3]Interval vectors and matrices are combined by the usual rules, each real operation replaced by its outward-rounded interval counterpart (Section 6), so every entry encloses all values the expression attains as x ranges over X. The endpoints of X are themselves doubles, and existence and embeddedness are certified over exactly this box.

Of the two ingredients only $F'(X)$ is an interval. The residual $F(\check{x})$ is a real vector, and the certificate cannot evaluate it exactly, so it computes an interval vector containing $F(\check{x})$ and evaluates the whole expression in interval arithmetic. Interval arithmetic is inclusion monotonic, so what the run returns is an interval vector containing the true Krawczyk image $K(\check{x}, X)$. Interior containment of the returned vector therefore gives interior containment of $K(\check{x}, X)$, which is what Theorem 3 asks for.

The two enclosures are computed by interval-valued forward-mode automatic differentiation [4] through the full analytic expression of G, deficits and developing map alike, in the outward-rounded interval arithmetic of Section 6. One implementation point matters and is easy to miss. The modulus was defined by developing, and for a flat configuration the development is well-defined. So the periods, and with them τ, do not depend on the order in which the triangles are unfolded. The box X, however, does not lie inside the flat locus. At a configuration with nonzero deficits the fan around a vertex misses closure by exactly the deficit (Figure 6, right). Developing around a homology loop then returns rotated by the deficits it encloses, there is no canonical period, and the value read off depends on the unfolding order. Every unfolding tree thus defines its own smooth extension of τ to non-flat configurations, and all extensions agree on the flat locus, where the ambiguity vanishes. For the theorem this is harmless. The certified zero is flat, and every extension reports the same modulus there. For the certificate we have to fix a choice and our matrix Y must approximate $F'(\check{x})^{-1}$, or $I - YF'(X)$ has no reason to be small. We therefore fix a single breadth-first unfolding tree, which also keeps the developed paths short and the interval widths small, because fewer unfolds along a loop means fewer operations to evaluate all enclosures through it, and take Y to be the float inverse [5] of the midpoint of the resulting interval Jacobian. In our experiments a finite-difference Y built from a worse unfolding fails to contract, while our choice yields a Y that contracts with $\|I - YF'(X)\|_\infty = 1.46 \times 10^{-5}$.

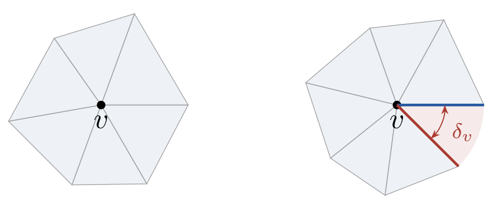

FIGURE 6. Developing the fan of triangles around a vertex v. On the left, the cone angle is exactly 2π, the fan closes up, and the development continues independently of the order of unfolding. On the right, the cone angle is $2\pi - \delta_v$, the fan misses closure by the deficit (red), and what the development reads off depends on the unfolding tree.

Theorem 4. *Let X be the box of radius $r = 10^{-9}$ around the center of Table 1 in the nine free coordinates, all other coordinates frozen at the center. Then G has exactly one zero $\bar{c}$ in X, a configuration that is exactly flat and has marked modulus exactly i, therefore a square torus.*

[4] A dual number carries an interval that encloses a value together with a vector of intervals that enclose its partial derivatives. Every arithmetic operation updates both parts by the chain rule in interval arithmetic, so evaluating G once over the box returns an enclosure of the residual $F(\check{x})$ and an enclosure of every entry of the Jacobian $F'(X)$ valid over the whole box.

[5] The inverse is computed by ordinary Gauss–Jordan elimination in double precision. Any invertible matrix is admissible as Y, and this one is close enough to $F'(\check{x})^{-1}$ to make the operator contract, as the reported contraction norm confirms.

Proof. Three certificates, run in the outward-rounded interval arithmetic of Section 6, give the residual enclosure $\|F(\check{x})\|_\infty \le 1.923 \times 10^{-13}$, the Jacobian enclosure over the box with entry widths at most 2.9×10^{-6}, and the Krawczyk containment $K(\check{x}, X) \subset \mathrm{int}(X)$ with slack 9.99×10^{-10} in every coordinate. By Theorem 3, F has exactly one zero in X, and a zero of F is a configuration with $G = 0$, exactly flat with marked modulus exactly i (Section 4). The containment holds verbatim at $r = 10^{-10}$ and $r = 10^{-11}$ as well, and the Krawczyk image has width below 1.3×10^{-12}, pinning the zero three orders of magnitude more tightly than the box that certifies it. The three runs are recorded as rows K1–K3 of the certificate table (Appendix A), with committed logs (`code/logs/certify-i-r1e-9.log`) and one-command reproduction (Appendix A). □

Here "exactly one" means one *in the slice*. Among the configurations that agree with the center on the fifteen frozen coordinates and differ by at most r on the nine free ones, $\bar{c}$ is the only zero of G. It is not a global statement, G vanishes on the whole 7-dimensional similarity orbit of $\bar{c}$, since a similarity leaves every cone angle and the modulus unchanged. Freezing fifteen coordinates isolates exactly one zero to certify.

Remark 2. The Krawczyk test needs F continuously differentiable on the whole box X, not just on the flat locus through it. It is, because F composes maps each smooth except at a degeneration. The developing map works from the edge lengths alone. It lays down the first triangle, then glues each next one to its parent along their shared edge and places its third vertex at the two prescribed edge lengths from the ends of that edge. The developed triangle is congruent to the one in the configuration, so it is non-degenerate exactly when that one is, and the placement is then a smooth function of the edge lengths, hence of the coordinates. The modulus reads the two period vectors off the developed net and forms $\tau = v_2/v_1$, smooth while the period v_1 is nonzero. And each deficit is 2π minus the triangle angles meeting at a vertex, again smooth on non-degenerate triangles. So F is smooth, in fact real-analytic, as long as the triangles stay non-degenerate and v_1 stays nonzero.

The certificate secures this as a by-product. The interval evaluation requires every triangle to stay non-degenerate and v_1 to stay nonzero over the whole box, aborting otherwise, so a completed run has already proved F analytic on X.

5. Embeddedness by separating axes

It remains to prove that $\bar{c}$ is embedded, and we prove more, that *every* configuration in the box X is embedded. The proof has two parts: interval certificates about pairs of triangles, and two lemmas of elementary geometry that convert the pair certificates into embeddedness over all of X. Both are stated in full.

5.1. **Embedding from pairwise intersections.**

The following lemma is elementary and, in $\mathbb{R}^3$, geometrically intuitive. We state and prove it because everything below is built on it.

Lemma 1 (a pairwise embedding criterion)**.** *Let c be a configuration such that (i) every face of T maps to a nondegenerate triangle, and (ii) for every pair of faces, the images intersect exactly in the image of their shared subcomplex, namely disjoint images for faces sharing no vertex, exactly the shared vertex's image for faces sharing one vertex, and exactly the shared edge's image for faces sharing an edge. Then c is an embedding.*

Proof. An affine map on a nondegenerate closed triangle is injective, so c is injective on each closed face by (i). Suppose $c(x) = c(y)$ with $x \in \sigma$, $y \in \sigma'$ for faces σ, σ'. The common value lies in $c(\sigma) \cap c(\sigma')$, which by (ii) is $c(\sigma \cap \sigma')$. Pick $z \in \sigma \cap \sigma'$ with $c(z) = c(x)$. Injectivity on σ

gives $x = z$, injectivity on σ' gives $y = z$, so $x = y$. A continuous injection of a compact space into a Hausdorff space is a homeomorphism onto its image, so c is an embedding. $\square$

5.2. **The certificates.** For T the $\binom{16}{2} = 120$ pairs split as 24 pairs sharing no vertex, 72 sharing exactly one vertex, and 24 sharing an edge (Figure 7).

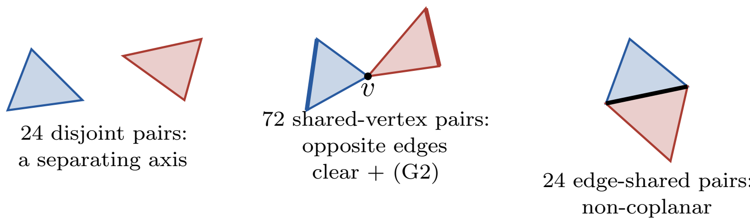

FIGURE 7. The pair census of the embedding proof and what is certified in each class.

All certificates below are validated computations over the box X. Every quantity is evaluated with the box coordinates as intervals, exact rational or outward-rounded, so a strict inequality certifies that inequality *simultaneously for every configuration in the box.* The labels (D), (V), (E) name the three classes of the pair census, and (G1), (G2) the two general-position conditions. The definitive run is in exact rational arithmetic (Section 6).

Theorem 5. *For every configuration in the box X of Theorem 4, all of the following hold, and consequently every configuration in X is embedded.*

- *(G1) Every face has nonzero area. For each of the 16 faces, some coordinate of the edge cross product is bounded away from 0 over X.*
- *(D) For each of the 24 vertex-disjoint pairs, some axis strictly separates the two triangles. The two projection intervals of the six vertices onto a certified axis are disjoint over X. The candidate axes are the two face normals, each face normal crossed with the edges of both triangles, and the nine edge-edge cross products.*
- *(V) For each of the 72 shared-vertex pairs, each triangle's opposite edge (the edge not containing the shared vertex) is strictly separated from the other closed triangle, by the same axis test applied to a segment and a triangle.*
- *(G2) For each of the 72 shared-vertex pairs, none of the four cross products $(a_i - v) \times (b_j - v)$ vanishes over X, where v is the shared vertex and a_i, b_j the other vertices of the two faces.*
- *(E) For each of the 24 edge-shared pairs, the tetrahedron spanned by the shared edge and the two opposite vertices has signed volume of one strict sign over X.*

The machine part is the validated evaluation itself, recorded as rows B1–B2 of the certificate table (Appendix A, logs `exact-embedding.log` and `certify-i-r1e-9.log` showing 16 faces, $24 + 72 + 24$ pairs, zero failures). Soundness of the separating-axis direction is immediate, since a strict projection gap on any axis proves disjointness. What remains is that (D), (V), (E), (G1), (G2) together give the hypotheses of Lemma 1. That is the content of the next two lemmas, after which the theorem is proved.

Remark 3. By the separating-axis theorem one of the axes in (D) separates any single disjoint configuration, so pointwise success is guaranteed, not lucky. Over the box the test needs one axis whose gap holds uniformly, which is more. The proof relies on neither fact. A strict projection gap on any single axis proves the pair disjoint over the whole box, and the evaluation exhibits one for each.

Lemma 2 (edge-shared pairs)**.** *Let A, B be nondegenerate closed triangles sharing the edge $[u, w]$, with the tetrahedral volume* $\det[\, w-u,\ a-u,\ b-u\,] \neq 0$*, where a, b are the opposite vertices. Then $A \cap B = [u, w]$.*

Proof. The volume condition says b is not in the plane of A, so the two planes are distinct. Both contain the line L through u and w, so they meet exactly in L, and $A \cap B \subseteq L$. A nondegenerate triangle meets the line through one of its edges exactly in that edge (the triangle lies in one closed half-plane bounded by L, and its third vertex is off L), so $A \cap L = [u, w] = B \cap L$. ☐

Lemma 3 (shared-vertex pairs)**.** *Let A, B be nondegenerate closed triangles sharing the vertex v, write $A = \operatorname{conv}(v, a_1, a_2)$ and $B = \operatorname{conv}(v, b_1, b_2)$, and suppose (i) the edge $[a_1, a_2]$ is disjoint from B and the edge $[b_1, b_2]$ is disjoint from A, and (ii) no $(a_i - v)$ is parallel to a $(b_j - v)$. Then $A \cap B = \{v\}$.*

Proof. $K = A \cap B$ is compact and convex and contains v. Suppose $K \neq \{v\}$ and let q be a point of K farthest from v, so $q \neq v$.

First suppose the planes of A and B are distinct. They share v, so they meet in a line $L \ni v$, and $K \subseteq S_A \cap S_B$ where $S_A = A \cap L$ and $S_B = B \cap L$ are segments containing v. On the side of v containing q, the point q is the nearer of the two segment endpoints. Say it is the endpoint of S_A (the other case is symmetric in A and B). An endpoint of S_A lies on the boundary of A, either on the opposite edge $[a_1, a_2]$, in which case $q \in [a_1, a_2] \cap B$ contradicts (i), or on a v-edge $[v, a_i]$. In the latter case v and q are two distinct points of $L \cap [v, a_i]$, so L is the line through the edge $[v, a_i]$, and $A \cap L = [v, a_i]$ exactly. The endpoint on the q side is then a_i itself, so $q = a_i \in [a_1, a_2] \cap B$, again contradicting (i).

Now suppose the planes coincide, so everything happens in one plane. The point q cannot lie in the interior of both triangles (a neighborhood of q would lie in K, containing points farther from v), so $q \in \partial A \cup \partial B$. Say $q \in \partial A$ (symmetric otherwise). If q is on the opposite edge $[a_1, a_2]$, then $q \in [a_1, a_2] \cap B$ contradicts (i) (this includes the endpoints $q = a_i$). So q lies on a v-edge, $q \in (v, a_i)$. If q were in the interior of B, the points $q + \varepsilon(a_i - v)$ would lie in A (the edge $[v, a_i]$ points along the ray from v through q) and in B for small $\varepsilon > 0$, and they are farther from v, a contradiction. So $q \in \partial B$ on the opposite edge $[b_1, b_2]$, contradicting (i) since $q \in A$, or on a v-edge of B, $q \in (v, b_j]$. Then $q \neq v$ lies on both rays from v through a_i and through b_j, so the rays coincide and $(a_i - v) \parallel (b_j - v)$, contradicting (ii). ☐

Proof of Theorem 5. All certificates hold over the whole box, so fix a configuration in X. Hypothesis (i) of Lemma 1 is certificate (G1). For hypothesis (ii), go through the three classes of pairs. Disjoint pairs have disjoint images by (D). Edge-shared pairs meet exactly in the shared edge by Lemma 2, whose volume hypothesis is certificate (E). Shared-vertex pairs meet exactly in the shared vertex by Lemma 3, whose hypothesis (i) is certificate (V) and hypothesis (ii) is certificate (G2). Lemma 1 then gives the embedding. ☐

Proof of Theorem 1. The zero $\bar{c}$ of Theorem 4 lies in X, and every configuration in X is embedded by Theorem 5. So $\bar{c}$ is an embedded, exactly flat polyhedral torus with $\hat{\tau} = i$, on the triangulation T, with 8 vertices. Vertex minimality is Schwartz's theorem that no paper torus has 7 vertices, together with the classical fact, also recorded in [10], that no torus triangulation has fewer. ☐

6. The certification model

This section explains what, exactly, the machine has proved and what we relied on. A *certificate* here is a finite computation in validated arithmetic, run by a module of the certificate suite inside `code/`, whose success implies a stated enclosure or strict inequality. The inputs are the committed center coordinates, the whole suite reruns in seconds with one command (Appendix A), and the logs are recorded.

6.1. **Two kinds of arithmetic.** The two halves of the proof make different demands. The existence half asks for a zero of the map G, which is not a rational function of the coordinates. Its deficits are sums of arccosines, and its developing map takes square roots of edge lengths. The embedding tests are polynomial: the separating axes are unnormalized cross products of edge vectors, the projections are dot products, the areas are cross products, and the volumes are determinants, with no square root or transcendental anywhere. The suite is split accordingly. Existence runs in *validated interval arithmetic*, in three independent implementations that all agree (Section 6.4). Embeddedness runs in *exact rational arithmetic*. A binary64 float is a rational with a power-of-two denominator, so the box X has exact rational endpoints, the entire reduction of Section 5 is evaluated in rational interval arithmetic with no rounding at any step, and every acceptance is a strict inequality between exact rationals. The embedding half of the theorem therefore rests on no floating-point assumption at all. The interval-double and Arb reruns of the same reduction are kept as redundancy (they agree with the exact result). The lemma layer is proved by hand from these machine-checked inputs: Lemmas 1–3, the Gauss–Bonnet reduction from 8 deficits to 7, the homology-basis property of the marking (Section 3), the orbifold-point argument of Section 2.3, and the assembly of Theorem 1.

6.2. **What the computation assumes.** Each part rests on exactly what it uses.

The handwritten interval arithmetic, the primary existence certificate, computes in double precision with directed outward rounding. Each endpoint, computed in round-to-nearest, is nudged one unit in the last place outward, and since a correctly rounded operation lands within half of that unit of the true value, the nudged interval strictly encloses it. It rests on three assumptions. First, IEEE-754 round-to-nearest correctness of $+, -, \times, \div$ on binary64 doubles. Second, a correctly rounded `math.sqrt`, which IEEE-754 mandates for the square root. Third, the correctness of `math.nextafter`, the single-unit step that implements the nudge. No library transcendental is assumed correct. The cosine is summed from its Maclaurin series in interval arithmetic with the explicit alternating-series tail bound, and the arccosine of a point is verified by an interval Newton step whose seed, our own transcription of fdlibm's `acos`, enters only as an unverified initial guess. The arccosine of an interval then uses the mean value form with a derivative enclosure built from $\sqrt{\ }$ and $\div$ alone. The Newton window is required to lie strictly inside $(0, \pi)$, where the cosine is injective, so the zero the step certifies is the principal arccosine and not another root of the same equation, and the enclosure of $\sin\theta = \sqrt{1-\cos^2\theta}$ is required strictly positive over the window, which is what legitimizes that substitution.

Arb and mpmath rerun existence in `python-flint` (Arb ball arithmetic, with Arb's rigorous cos, arccos, $\sqrt{\ }$) and in `mpmath.iv` at 200 bits (with an interval arccosine built by bisection against the verified interval cosine, so it relies on no library transcendental either). Each rests on its library's rigorous arithmetic and nothing else.

The exact rational arithmetic, which carries the embedding, the asymmetry certificate, and the rational witness of Appendix B, rests on exact integer arithmetic alone.

Python itself does not specify its floating point, so the IEEE-754 assumptions above are about the platform CPython runs on. A platform that got the arithmetic wrong would break the suite's acceptance gates, which reproduce the handwritten enclosures bit for bit against a frozen reference run and require the three backends to agree. And the mpmath, Arb, and exact rational reruns rest on no platform floating point at all.

6.3. **Interval automatic differentiation.** Automatic differentiation evaluates a function and its derivatives together, by carrying derivative information alongside every intermediate value. Each quantity in the evaluation of G is a *dual number*, an interval that encloses a value together with a short vector of intervals that enclose its nine partial derivatives with respect to the free coordinates. The rules of calculus become rules on dual numbers. A sum adds both parts. A product uv carries the value uv and the derivative vector $u'v + uv'$, each entry computed in interval arithmetic. A square root or an arccosine carries its usual derivative, evaluated on the enclosed value. Evaluating G at the center encloses the residual $F(\check{x})$, and evaluating it with the box coordinates fed in as dual numbers encloses every entry of the Jacobian $F'(X)$ over the whole box. These are the two inputs the Krawczyk operator needs, and since every step is an interval operation, the true residual and Jacobian are guaranteed to lie within the intervals returned.

6.4. **Independent cross-validation.** So that the result does not rest on one implementation, the suite certifies everything at least twice, in arithmetics of different kinds. Its one command enforces the agreements as hard acceptance gates, and any failure exits nonzero.

(1) *Existence, three ways*. The handwritten backend, Arb, and `mpmath.iv` each certify the Krawczyk containment on both tori, over the identical box, and agree on the residual, the modulus, and the interior slack to far below the certified margin. On the square torus the residual bounds are 1.93×10^{-13} (handwritten, at the outward-rounded box endpoints) and 9.98×10^{-16} (Arb and mpmath, at higher working precision), the modulus reproduces within 6×10^{-16} of i, and all slacks are $\approx 10^{-9}$.
(2) *Embedding, three ways*. The exact rational reduction is the proof. The interval-double and Arb reruns must certify the identical census with zero failures, and they do.

6.5. **Reproducibility as evidence.** The full chain is deterministic and reruns from one command in about six seconds (Appendix A). Every certificate regenerates its log, and the acceptance gates above must all pass. The suite depends on the Python standard library, `python-flint`, and `mpmath` alone, and numpy is deliberately absent, because the dependency policy admits only arithmetic whose rounding is specified.

7. Relation to Schwartz's argument

Our argument uses the same ingredients as Schwartz's: a high-precision near-solution, a square system of cone-angle equations, a well-conditioned Jacobian, and an effective existence theorem. What it adds is a modulus his argument cannot reach. His order-2 symmetry $\rho(x, y, z) = (-x, -y, z)$ is the elliptic involution every flat torus carries. It halves his system to 3×3 but constrains nothing about the modulus, so their pup tents sweep an *open* set of moduli, which Doyle and Schwartz [11] prove is the interior of their bi-cusped fundamental domain. It contains every reflection-asymmetric flat torus, almost all flat tori. The square torus is reflection-symmetric, the measure-zero case that open set excludes. Realizing $\hat{\tau} = i$ on the same triangulation as the pup tents, imposing no symmetry, pinning the modulus with the two equations $\tau = (0, 1)$, and validating existence and embeddedness in full interval arithmetic, is therefore a new existence statement. The validation is where the arguments diverge most. His second-derivative bound (Lemma 3.4 of [10]) is proved by hand and his robust-embedding certificate runs in exact integer arithmetic over a box, but his existence step assumes the correctness of two high-precision floating-point computations, the residual and the Jacobian at the center. We enclose residual, Jacobian, and the whole embedding suite in validated arithmetic over the box and apply Krawczyk's test. He also never needs the edge-shared pairs, which Lemma 3.2 of [10] shows are implied. We certify them directly.

8. Experimental findings and open questions

The certified square torus is one point on one triangulation. Our searches explored beyond it. The findings below are numerical, checked but not certified.

The square modulus on the other types. Of the seven 8-vertex torus triangulations, we realized $\hat{\tau} = i$ on five, among them the valence-regular type certified here. Of the remaining two, one has a vertex of valence 3, where a flat metric forces the three adjacent faces to be planar or folded. Folded contradicts embeddedness, and planar effectively erases the vertex, leaving a 7-vertex paper torus, impossible by [10]. The last, with degree sequence $(4, 5, 5, 6, 7, 7, 7, 7)$, resisted every search. We found no flat torus on it at any modulus, not even numerically, and we conjecture that none exists.

The rest of the excluded locus. The moduli that near universality excludes form a measure-zero locus inside the reflection-symmetric family, and this paper certifies both of its corners, the square torus $\hat{\tau} = i$ here, and the hexagonal torus $\hat{\tau} = e^{i\pi/3}$ on a different triangulation (Appendix C). The rest, the rectangular tori $\hat{\tau} = iY$ for $Y > 1$ and the rhombic tori of aspect ratio greater than $\sqrt{3}$, is left open. The rhombic arc between the two orbifold points is not, since the pup tents already realize it [11]. Certified realizations on the two open rays are a natural next target. Work on them is under way, and we are confident that both can be closed.

Symmetric realizations. Our realization of the square torus is asymmetric by construction. Whether a maximally symmetric 8-vertex realization of $\hat{\tau} = i$ exists, and whether imposing a reflection would pin the modulus for free the way the pup tents' rotation cannot, is open.

What the method gives, and what it does not. The certificates are pointwise. Each run pins one modulus and proves that one paper torus exists, and nothing in the argument produces a family. The method is not special to the two points certified here, and it closes at other moduli and on other triangulations, but it yields no statement of the form "every flat torus of this kind is realized". That is the difference from [8], whose single triangulation admits every flat torus, and from [11], whose pup tents sweep an open set. We were after the two most symmetric points, which the general arguments do not reach, and for those a pointwise certificate is what the problem calls for.

Appendix A. Certificates

Each row lists the certified fact, the method, the module of the certificate suite (under `code/`), the committed log (under `code/logs/`), and where it enters the paper. The rows are grouped by role: K for the Krawczyk existence chain, B for the embedding certificates over the box, U for the untrusted float constants, C for the independent cross-validations, and W for the exact-rational facts about the committed center itself. In the method column, AD abbreviates automatic differentiation, SAT the separating-axis test, SVD the singular value decomposition, and SNF the Smith normal form. Every row regenerates from the one command below. The committed input of every square-torus certificate is `code/data/square-i-type7.csv`. The hexagonal companion's rows, cited in Appendix C, run from `code/data/rho-type3.csv` in the same command (logs `certify-rho-r1e-9.log`, `model-checks-rho.log`).

Fact	Method	Module	Log	Enters
K1 residual enclosure $\\|F(\check{x})\\|_\infty \le 1.923 \times 10^{-13}$	interval AD	`existence/handwritten/`	`certify-i-r1e-9.log`	Thm 4
K2 Jacobian enclosure over X, widths $\le 2.9 \times 10^{-6}$	interval AD over the box	same	same	Thm 4
K3 Krawczyk containment at $r = 10^{-9}, 10^{-10}, 10^{-11}$	Krawczyk operator	same	`certify-i-r*.log`	Thm 4
B1 the 120 pair separations (D), (V), (E) over the box	exact rational SAT and signed volume; interval-double and Arb reruns	`embedding/exact.py`	`exact-embedding.log`, `certify-i-r1e-9.log`	Thm 5
B2 general position (G1), (G2) over the box	exact rational cross products	`embedding/exact.py`	same	Thm 5
U1 float constants (κ and the free-coordinate set)	float AD midpoints, Jacobi SVD	`constants.py`	`square-constants.log`	§4
C1 model fidelity (topology, H_1 basis, flatness)	exact integers (SNF)	`model/topology.py`	`model-checks.log`	App. C
C2 residual, modulus, embedding rerun	`mpmath.iv`, 200 bits	`existence/mpmath/`	`intval-crosscheck.log`	§6.4
C3 Krawczyk rerun	`mpmath.iv` interval AD	`existence/mpmath/`	`krawczyk-crosscheck.log`	§6.4
C4 existence, modulus, embedding in Arb	`python-flint` balls	`existence/arb/`, `embedding/arb.py`	`arb-crosscheck.log`	§6.4
W1 exactly-embedded rational witness	exact rationals (`Fraction`)	`witness.py`	`witness-i.log`	App. B
W2 asymmetry of both centers (no isometry permutes the vertices)	exact rational distance matrix	`exact_embedding.py`	`exact-embedding.log`	§3

The free-coordinate set $\{z_3, z_0, y_1, y_5, z_6, z_4, x_7, z_5, y_0\}$ is recorded in the K and B logs, and row U1 recomputes it from scratch by the column pivot of Section 4.

The suite is the self-contained folder `code/`, published as a standalone repository at

`https://github.com/FabianLander/square-hex-paper-tori`

whose root is the folder the table above calls `code/`. From that folder, one command reruns the whole proof:

```
python3 run.py
```

It regenerates every row above for both tori, writes the logs to `code/logs/`, and enforces the agreements of Section 6.4 as hard acceptance gates, exiting nonzero if any fails. The whole chain finishes in about six seconds on one laptop core, and `--verbose` echoes each log as it is produced. The environment is Python 3.13.12 with `mpmath` 1.3.0 and `python-flint` 0.8.0. The candidate configuration came from the exploratory repository `low-vertex-flat-tori` (search and polish, no validated code).

Appendix B. An exactly-embedded rational witness

The certified flat torus $\bar{c}$ has irrational coordinates (it is a zero of transcendental constraints), so it cannot be handed over as a finite list of numbers. What can be handed over is Table 1 itself, and it is better than a mere approximation. Read as exact rational numbers (a binary64 float is a rational with denominator a power of two, and the largest denominator in the table is 2^{57}), the committed center is **exactly embedded**, verified in exact rational arithmetic. The verification (`code/witness.py`, log `code/logs/witness-i.log`) reruns the entire reduction of Section 5 over $\mathbb{Q}$, exhibiting a strictly separating axis for each of the 24 disjoint pairs, the opposite-edge tests and the four non-collinearity cross products for each of the 72 shared-vertex pairs, a nonzero exact volume for each of the 24 edge-shared pairs, and nonzero exact area for all 16 faces. Every acceptance is a strict inequality in `Fraction` arithmetic. There is no clipping step and no skip path, so the zero-area-touch pitfall cannot arise here by construction. Anyone can re-check embeddedness of this configuration with rational arithmetic alone.

The witness is not exactly flat. A 40-digit evaluation places its deficits below 10^{-15} and its marked modulus within 6×10^{-16} of i. The exactly flat object is its neighbor $\bar{c}$ in the Krawczyk box, within 10^{-9} in every coordinate. The same log records two more facts. Newton-polishing the center to 40 digits moves the nine free coordinates by at most 1.21×10^{-15} each, and the corrections all round to zero at denominator 10^{12}, so the committed center is its own best rational approximation at that scale. The ancillary file `code/logs/witness-i.txt` ships the exact rational coordinates.

Appendix C. The hexagonal companion

This appendix proves Theorem 2. The argument is that of the square case, Sections 4 and 5, carried out on the other triangulation and with the modulus pinned to $\rho = e^{i\pi/3}$. The embedding criterion and its two supporting lemmas (Lemmas 1–3) and the certification model of Section 6 are independent of the combinatorics and of the modulus, and apply verbatim. Only the triangulation, the marking, the modulus target, and the certificate numbers change.

The triangulation. The realization lives on the triangulation T', type #3 in the enumeration of [12, 13], whose 16 faces in slot order are

$$012,\ 031,\ 024,\ 043,\ 152,\ 136,\ 145,\ 174,$$
$$167,\ 264,\ 257,\ 276,\ 347,\ 356,\ 375,\ 465.$$

Its degree sequence is $(4, 6, 6, 6, 6, 6, 7, 7)$, where vertex 0 has valence 4, vertices 1 and 4 valence 7, and the rest valence 6. It is not the valence-regular type of the square case.

The marking. The two marking loops are the red loop $1 \to 4 \to 0 \to 1$ and the blue loop $1 \to 5 \to 6 \to 1$. They are simple closed curves and meet only at vertex 1. The seven faces incident to that vertex are $012, 031, 152, 136, 145, 174, 167$, so its neighbors close up into the link cycle $(2, 0, 3, 6, 7, 4, 5)$. The red loop leaves vertex 1 along the edges to 4 and 0, the blue along the edges to 5 and 6, and in the link cycle these two pairs alternate. The two loops therefore cross exactly once, and the argument of Section 3 applies verbatim, so the marking is a homology basis.

The orbifold point at ρ. Because $\rho = e^{i\pi/3}$ is a fixed point of the order-3 element $\tau \mapsto (\tau - 1)/\tau$ of $\mathrm{PSL}(2, \mathbb{Z})$ (the matrix $\left(\begin{smallmatrix} 1 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, whose cube is $-I$), it is an orbifold point of order 3, and the reduced representative $\hat{\tau}$ is not a smooth coordinate there. As at i, we pin the raw modulus, now by

$$\tau_{\mathrm{re}}(c) = \tfrac{1}{2}, \qquad \tau_{\mathrm{im}}(c) = \tfrac{\sqrt{3}}{2}.$$

A zero of the constraint map then has raw modulus ρ. The target $\sqrt{3}/2$ is irrational. It is enclosed outward in the certificate exactly as 2π is in the deficits (Section 4), so the certified zero has $\tau = \rho$ exactly.

The certified configuration. The certified center is the configuration of Table 2, found by the same numerical search-and-polish as in the square case. Its eight cone angles equal 2π to within 2.7×10^{-15}, so it is flat to full double precision, and as in the square case no symmetry is imposed on the construction.

v	x_v	y_v	z_v
0	−0.0052919576658824819	0.25189213380754671	−0.19636317590069088
1	−0.10330037049737666	0.097954858850342955	0.23022989757571455
2	0.088435194516004872	−0.042881936616869155	−0.16602959156361255
3	0.10713507949378637	−0.03807120945413147	−0.19492235829557528
4	−0.15821722275466177	0.26928131058897076	−0.057266978045909966
5	0.099801133842719214	0.17571318949877185	0.042053203465396087
6	−0.026706140635424078	0.3003490487315576	−0.18681063266426795
7	−0.004032608759816151	−0.14181859043688222	−0.02715724967818926

TABLE 2. The certified hexagonal center, the eight vertex positions to full double precision, machine-emitted from the committed data file `code/data/rho-type3.csv`. The exactly flat torus of modulus exactly $\rho = e^{i\pi/3}$ of Theorem 2 lies within 10^{-9} of it, coordinate by coordinate.

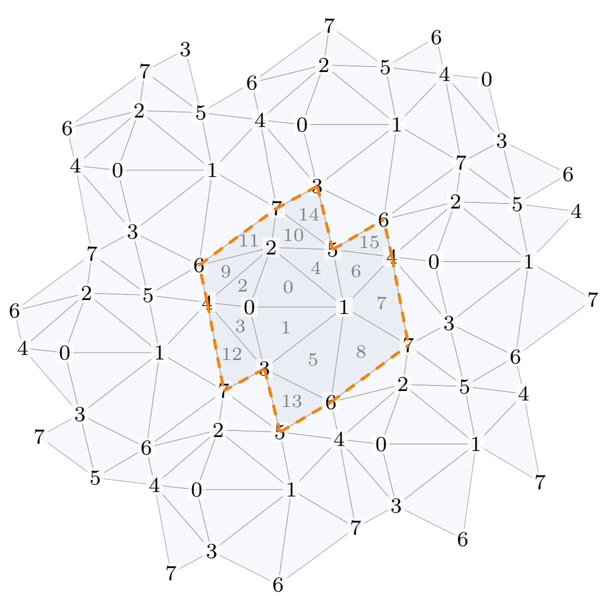

FIGURE 8. The triangulation T', shown through the universal cover of the certified flat metric. The sixteen developed triangles and their translates by the period lattice tile the plane, with the boundary of our unfolding dashed and the faces numbered in slot order inside it. Every vertex appears with its full fan of faces, so the valences $(4, 6, 6, 6, 6, 6, 7, 7)$ can be read off.

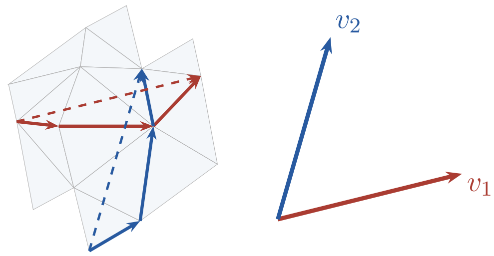

FIGURE 9. The development of the certified hexagonal configuration with the two marking loops (red for $1 \to 4 \to 0 \to 1$, blue for $1 \to 5 \to 6 \to 1$) and their holonomy vectors, drawn dashed on the net. The inset repeats the two vectors from a common origin, to the same scale. v_2 is v_1 turned by $60°$, so $\tau = v_2/v_1 = \rho = e^{i\pi/3}$ to 15 digits.

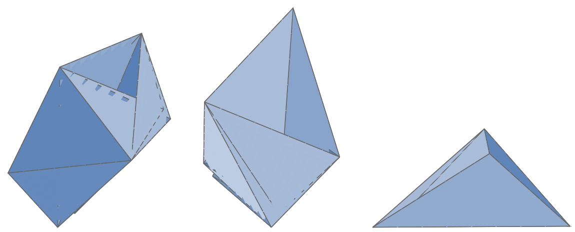

FIGURE 10. The certified embedded hexagonal paper torus from three angles, orthographic projections of the configuration of Table 2.

The certificates. Column-pivoted Gram–Schmidt on the Jacobian at the center selects the nine free coordinates $\{ y_4,\ z_1,\ z_3,\ z_2,\ y_6,\ x_5,\ y_7,\ z_6,\ y_3 \}$. Over the box X of radius $r = 10^{-9}$ around the center in these coordinates, the certificates of Section 6 give the residual enclosure $\|F(\check{x})\|_\infty \le 2.747 \times 10^{-13}$, the Jacobian enclosure over the box with entry widths at most 4.17×10^{-5}, and the Krawczyk containment $K(\check{x}, X) \subset \operatorname{int}(X)$ with interior slack 9.995×10^{-10} in every coordinate (contraction norm $\|I - YF'(X)\|_\infty = 7.52 \times 10^{-5}$, Krawczyk image width below 8.9×10^{-13}). By Theorem 3, F has exactly one zero in X, a configuration that is exactly flat and has marked modulus exactly ρ.

The embedding suite of Theorem 5 runs over the same box with the pair census of T', which splits as 21 vertex-disjoint, 75 shared-vertex, and 24 edge-shared pairs ($21 + 75 + 24 = 120$), and every test passes. The 16 faces are nondegenerate (G1), every disjoint pair is strictly separated (D), every shared-vertex pair passes the opposite-edge and non-collinearity tests (V), (G2), and every edge-shared pair has one-signed tetrahedral volume (E). All pass, with zero failures, so every configuration in X is embedded by Lemmas 1–3.

Proof of Theorem 2. The Krawczyk zero lies in X, and every configuration in X is embedded. So the zero is an embedded, exactly flat polyhedral torus on T' with 8 vertices, 24 edges, and 16 faces, and reduced representative $\hat{\tau} = e^{i\pi/3}$. □

The certificates are produced by the suite of Appendix A, whose one command runs the handwritten existence and embedding chain on T' (log `code/logs/certify-rho-r1e-9.log`), the exact-integer model check (log `code/logs/model-checks-rho.log`), and the exact-rational embedding with the asymmetry certificate (log `code/logs/exact-embedding.log`), and its acceptance gates cover the hexagonal torus alongside the square one.

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